

Detecting Bot-Controlled Accounts on Social Media

This project explores machine learning techniques to detect bots on Instagram and Twitter. The growing bot threat manipulates opinions, spreads fake news, and commits fraud. Traditional methods struggle, so adaptive models using behavioral and content features are needed.

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Background & Motivation



Vulnerabilities in Social Media

Bots infiltrate platforms, mimicking humans for manipulation.



Limitations of Rule-Based Systems

Traditional detection struggles with evolving bot behaviors.



Need for ML-Based Detection

Automated, adaptive, interpretable methods are essential.

Problem Statement & Objectives

Challenge

Detect bots on diverse platforms with distinct data types.

- Instagram focuses on visual content.
- Twitter relies on text-heavy data.

Approach and Goals

- Extract platform-specific behavioral and content features.
- Train Random Forest classifiers for robust prediction.
- Use TF-IDF for nuanced Twitter text analysis.
- Evaluate with Accuracy, Precision, Recall, F1, ROC-AUC.



Significance of the Project

Growing Threat

Social bots increasingly influence politics, scams, and misinformation.

Cross-Platform Complexity

Detection needs tailored strategies for Instagram and Twitter.

Impactful Contribution

Combines text processing with behavioral analytics.

Future Potential

Foundation for real-time, cross-platform bot detection systems.



Dataset: Cresci 2017

Instagram Data

Profile metadata and activity logs.

Twitter Data

Tweets, metadata, and network information.

Labels

Bot and genuine accounts labeled for supervised learning.

Data Preprocessing

1

Clean and Normalize

Handle missing values; apply Min-Max scaling on features.

2

Label Encoding

Encode targets: 0 for humans, 1 for bots.

3

Twitter-Specific Cleaning

Remove links, mentions, and punctuation from tweets.

4

Train-Test Split

80/20 stratified split ensures balanced evaluation.





Feature Engineering



Instagram Features

- Follower-to-following ratio
- Post frequency and bio length
- Media counts per user



Twitter Features

- Retweet and reply ratios
- Hashtag and URL mentions
- Average tweet timing (intertime)

TF-IDF: Text Feature Extraction



TF (Term Frequency)

Counts word occurrence in user tweets



IDF (Inverse Document Frequency)

Downweights common words across users



TF-IDF Calculation

Highlights words unique to bots or humans



Application

Captures spammy/repetitive bot language on Twitter

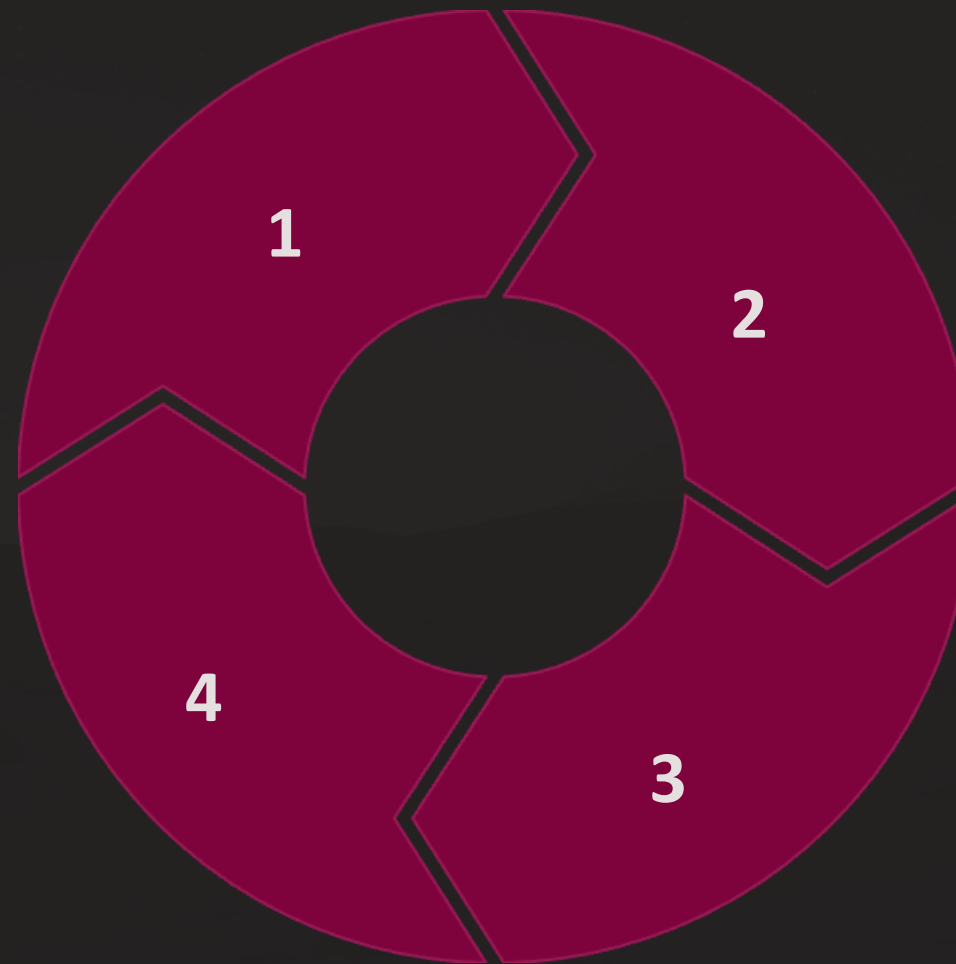
Random Forest Classifier

Ensemble Learning

Combines predictions from multiple decision trees.

Benefits

Reduces overfitting and improves generalization on metadata and text.



Random Subsets

Trees trained on random data and feature subsets.

Majority Voting

Final prediction based on majority vote of trees.

How Decision Trees Work

Node Condition

Tests attributes like follower ratio > 3?

Data Splitting

Branches data to child nodes based on condition

Leaf Nodes

Assign class label: Bot or Human

Random Forest

Hundreds of trees, majority voting for final decision



Training and Evaluation

Cross-Validation

5-fold to ensure model generalization

Hyperparameters

Tuned trees=300, depth=30, criterion=gini

Platform-Specific Models

Separate models for Instagram and Twitter

Metrics

Accuracy, Precision, Recall, F1, ROC-AUC



Instagram Detection Results

94.2%

Accuracy

93.7%

Precision

94.5%

Recall

96.3%

ROC-AUC

Indicators: Low follower/following ratio, unusual posting frequency

Twitter Detection Results

98.8%

Accuracy

98.5%

Precision

99.0%

Recall

99.2%

ROC-AUC

TF-IDF features greatly enhance detection of spammy language



Comparison & Analysis

1

Twitter

Better due to rich textual analysis using TF-IDF

2

Instagram

Relies mainly on behavior-based metadata

3

Twitter Bots

Spammy language patterns detected

4

Instagram Bots

Abnormal post timing, URL usage, username length

Conclusions and Future Scope

Conclusion

Platform-aware ML with behavioral + content features effective

Future Deep Learning

Incorporate LSTM, BERT, and Graph Networks

Real-time Detection

Deploy live bot detection pipelines

Explainability

Integrate Explainable AI for model transparency

